

Cover the answer · attempt aloud · check · re-test the misses tomorrow

CO ARITHMETIC — SV, EF, FICK

6 CARDS

1 Write the four core cardiac-output equations.

$$CO = SV \times HR \cdot SV = EDV - ESV \cdot EF = SV / EDV \cdot \text{Fick } CO = VO_2 / (CaO_2 - CvO_2).$$

2 EDV 125 mL, ESV 75 mL, HR 100. Compute SV, EF, and CO.

$$SV = 50 \text{ mL} \cdot EF = 50/125 = \mathbf{40\%}$$
 (HFrEF range) · $CO = 50 \times 100 = \mathbf{5 \text{ L/min}}$.

3 EF cutoffs for HFrEF, HFmrEF, and HFpEF?

$$HFrEF < 40\% \cdot HFmrEF \text{ 40--49\%} \cdot HFpEF \geq 50\%.$$
4 VO_2 250 mL/min and an A-V O_2 difference of 5 mL/dL. Calculate CO by Fick.
$$CO = 250 / (5 \text{ mL/dL} \times 10 \text{ dL/L}) = 250 / 50 = \mathbf{5 \text{ L/min}}$$
.5 Why is coronary sinus O_2 saturation the lowest in the body?

Resting myocardium extracts >70% of delivered O_2 (highest of any tissue). With no extraction reserve, the leaving (sinus) blood is profoundly desaturated and any extra demand has to be met by $\uparrow$ flow.

6 Two physiologic determinants of pulse pressure?

Stroke volume (a bigger SV produces a bigger surge) and **aortic compliance** (a stiffer aorta amplifies the surge into a bigger pressure swing).

HEMODYNAMICS — OHM, POISEUILLE, EDEMA

4 CARDS

7 State Ohm's law adapted to the systemic circulation.

$$MAP - CVP \approx CO \times SVR.$$
 Useful for parsing how vasopressors / vasodilators / diuretics each move the equation.

8 In Poiseuille's law, which variable controls resistance most strongly?

Vessel radius (r^4). Halving the radius raises resistance 16-fold. Arterioles modulate BP because small radius changes have huge effects.

9 Four Starling-equation mechanisms of peripheral edema?

$$\uparrow \text{ capillary hydrostatic (HF, venous obstruction)} \cdot \downarrow \text{ capillary oncotic (nephrotic, cirrhosis, malnutrition)} \cdot \uparrow \text{ capillary permeability (sepsis, burns, anaphylaxis)} \cdot \text{lymphatic obstruction (filariasis, post-axillary node dissection)}.$$

10 Clinical reading of a wide pulse pressure in an elderly patient with no other symptoms?

Isolated systolic hypertension from age-related loss of aortic compliance. High SBP, normal DBP, wide PP.

11 Define preload, afterload, and contractility in one line each.

Preload $\equiv$ LV end-diastolic volume (depends on venous return). **Afterload** $\approx$ aortic diastolic pressure (the pressure LV must overcome to open AV). **Contractility** = intrinsic force at a given preload (independent of EDV).

12 Molecular basis of the Frank-Starling relationship?

$\uparrow$ EDV stretches sarcomeres toward the optimal length ($\sim 2.2 \mu\text{m}$), maximizing actin-myosin overlap and length-dependent activation of troponin C $\rightarrow$ more force per beat $\rightarrow \uparrow$ SV.

13 Five inputs that raise myocardial contractility?

Catecholamines (β_1 agonism), digoxin (Na/K ATPase block $\rightarrow \uparrow$ cytosolic Ca via NCX), $\uparrow$ extracellular Ca, $\downarrow$ extracellular Na (drives NCX), thyroid hormone.

14 Five inputs that lower myocardial contractility?

β -blockers, non-DHP CCBs, acidosis, hypoxia / ischemia, HFrEF.

15 Why does hypovolemic shock improve with a saline bolus on a Frank-Starling basis?

Volume restoration $\rightarrow \uparrow$ EDV $\rightarrow \uparrow$ end-diastolic sarcomere length $\rightarrow \uparrow$ force per beat $\rightarrow \uparrow$ SV.

16 How does digoxin raise contractility at the cellular level?

Inhibits sarcolemmal Na/K ATPase $\rightarrow \uparrow$ intracellular Na $\rightarrow$ reduces the Na gradient that drives NCX $\rightarrow$ less Ca extrusion $\rightarrow \uparrow$ cytosolic Ca for cross-bridge cycling.

BEDSIDE MANEUVER DYNAMICS

5 CARDS

17 Murmur that gets louder with Valsalva strain phase?

HCM (and mitral valve prolapse). $\downarrow$ Preload worsens the dynamic LVOT obstruction; almost every other murmur softens.

18 Murmur that gets softer with sustained handgrip?

HCM and aortic stenosis (handgrip $\uparrow$ afterload, which fills the LV cavity / opposes ejection $\rightarrow$ less obstruction in HCM, less LV-aortic gradient in AS).

19 Which murmurs intensify with inspiration, and what is the sign called?

Right-sided murmurs (TR, TS, PR, PS) — **Carvallo sign**. Inspiration $\uparrow$ venous return to the right heart.

20 How does the MVP click move with Valsalva?

Click moves **earlier** in systole — $\downarrow$ preload $\rightarrow$ smaller LV cavity $\rightarrow$ redundant leaflets prolapse sooner.

21 Quick way to distinguish HCM from AS at the bedside?

Valsalva makes HCM *louder* but AS *softer*; squat / handgrip do the opposite. HCM is the murmur that "moves the wrong way."

22 Name the four corners of the PV loop in order.

1 MV closes (S1) · 2 AV opens · 3 AV closes (S2) · 4 MV opens (opening snap if MS). 1→2 isovolumetric contraction; 2→3 ejection; 3→4 isovolumetric relaxation; 4→1 filling.

23 What does loop width represent?

Stroke volume (EDV – ESV).

24 What does the slope of the end-systolic pressure-volume relationship (ESPVR) represent?

Contractility. A steeper line = higher inotropy (dobutamine, exercise). A flatter line = depressed contractility (β -blocker, HFrEF).

25 Loop changes after a saline bolus?

Loop shifts **right** and becomes **wider**: $\uparrow$ EDV, $\uparrow$ SV, ESV unchanged. Peak LV pressure rises slightly.

26 Loop changes with phenylephrine?

$\uparrow$ Afterload $\rightarrow$ loop becomes **taller and narrower**: peak LV pressure rises, $\uparrow$ ESV, $\downarrow$ SV. Reflex bradycardia (intact baroreflex) often accompanies.

27 Loop changes with dobutamine?

$\uparrow$ Contractility $\rightarrow$ ESPVR steeper $\rightarrow$ $\downarrow$ ESV, $\uparrow$ SV. EDV unchanged.

28 Loop changes with nitroglycerin?

Venodilation $\rightarrow$ $\downarrow$ preload $\rightarrow$ $\downarrow$ EDV, $\downarrow$ end-diastolic sarcomere length, $\downarrow$ SV, $\downarrow$ peak pressure. Narrow loop. Lowers wall stress $\rightarrow$ antianginal.

29 Loop changes with nitroprusside?

Balanced venous + arterial dilation $\rightarrow$ loop shrinks at both ends but **SV is preserved**. Drug of choice in acute HF with HTN.

30 PV-loop signature of a long-standing AV fistula?

$\uparrow$ Preload + $\downarrow$ afterload $\rightarrow$ wide loop with markedly $\uparrow$ SV. Chronic state = high-output cardiac failure.

31 On a normal PV loop, which point corresponds to a mitral opening snap?

Point 4 — the mitral valve opens at the end of isovolumetric relaxation.

32 Which JVP wave disappears in atrial fibrillation?

The **a wave** (no organized atrial systole).

33 What ECG / arrhythmia produces cannon a waves?

Any state where atria contract against a closed tricuspid valve — classically **complete (3°) heart block** with AV dissociation, also junctional rhythms.

34 What does a giant v wave signal?

Tricuspid regurgitation — regurgitant jet adds to RA pressure during ventricular systole. Often accompanied by a pulsatile liver.

35 Steep y descent vs blunted y descent on JVP?

Steep y descent → **constrictive pericarditis** (sudden ventricular filling abruptly halted by the rigid pericardium).
Blunted y descent → **tamponade** (ventricular filling impaired throughout diastole).

36 On mitral inflow Doppler, what do the E and A waves represent?

E wave = early passive ventricular filling. **A wave** = late atrial-contraction filling.

37 What is happening at the time of S3?

Sudden deceleration of rapid passive filling against a volume-overloaded or non-compliant LV (end of the E wave).
Normal under 40 yo; pathologic in adult HF.

HEART SOUNDS & S2 SPLITS

5 CARDS

38 Mechanism of S1 and S2?

S1 = MV + TV closure (start of systole). S2 = AV + PV closure (start of diastole; A2 then P2 normally).

39 Why is S2 a fixed split in an ASD?

RA and LA pressures are tied across the shunt all the time, so the inspiratory delay of P2 is present in both phases of respiration → A2–P2 separation is constant.

40 Two classic causes of paradoxical S2 splitting?

LBBB and **severe aortic stenosis** — both delay A2 past P2; split widens on expiration, unifies on inspiration.

41 Mechanism and significance of S4?

Atrial contraction forcing blood into a stiff, non-compliant LV. In adults, S4 is almost always pathologic — HCM, HFpEF, chronic HTN with concentric LVH, AS.

42 Best way to listen for S3 / S4?

Bell of the stethoscope, patient in left lateral decubitus, apex. Low-pitched diastolic sounds need a bell + the leftward shift to be heard.

43 Classic triad of severe aortic stenosis, with mean survival once symptomatic?

Angina (~5 y), syncope (~3 y), heart failure (~2 y). Once symptoms appear, mortality climbs sharply; valve replacement is indicated.

44 Peripheral signs of chronic aortic regurgitation?

Water-hammer (Corrigan) pulse, wide pulse pressure, low DBP, de Musset (head bobbing), Quincke (nail-bed pulsations), Müller (uvular pulsations), Duroziez (to-and-fro femoral bruit), Traube (pistol-shot femoral), Hill (popliteal SBP > brachial by >60 mmHg).

45 Why does the LV pressure tracing in mitral stenosis look normal?

The obstruction is upstream of the LV. The LV is just *under-filled*; LVEDP, afterload, and contractility are normal. The LA is the chamber under pressure — PCWP ↑ with normal LV pressures.

46 Cath signature of mitral stenosis?

A persistent LA – LV diastolic gradient (elevated PCWP, normal LV pressures). Severity tracks the size of the gradient.

47 Why does the PV loop in mitral regurgitation lack a true isovolumetric contraction phase?

As soon as LV pressure rises at S1, the incompetent MV leaks blood back into the LA before the AV opens — volume falls continuously instead of staying flat.

48 Most common cause of acute severe MR 3–5 days after an inferior STEMI?

Rupture of the **posteromedial papillary muscle** (single PDA blood supply). Presents as flash pulmonary edema and cardiogenic shock; needs emergent surgery.

49 Auscultatory profile of mitral valve prolapse?

Mid-systolic click + late-systolic murmur at the apex. Click moves earlier with Valsalva (↓ preload → redundant leaflets prolapse sooner).

50 How does endocarditis from IV drug use typically present?

Right-sided valve, most often the tricuspid; organism most often *Staphylococcus aureus*. Holosystolic murmur at LLSB louder on inspiration (Carvallo), septic pulmonary emboli, fever, peripheral stigmata.

CARDIOMYOPATHY PHENOTYPES

4 CARDS

51 Concentric LVH vs eccentric LVH — cause and sarcomere geometry?

Concentric = pressure overload (HTN, AS); sarcomeres added *in parallel*; thick walls, normal/small cavity. **Eccentric** = volume overload (AR, MR, AV fistula, dilated CMP); sarcomeres added *in series*; dilated cavity.

52 HCM genetics — mode of inheritance and two most-common mutated genes?

Autosomal dominant, incomplete penetrance. Most common: β -myosin heavy chain (**MYH7**) and myosin-binding protein C (**MYBPC3**).

53 Echo profile distinguishing HCM from athlete's heart?

HCM: small cavity, asymmetric septal hypertrophy, impaired diastolic relaxation, S4. Athlete's heart: normal-to-mildly-dilated cavity, symmetric mild hypertrophy, normal diastole, reversible with detraining.

54 Why is exertional sudden cardiac death the feared complication of HCM?

Exercise drops preload (catecholamine venodilation, shorter diastole) and raises contractility, both worsening dynamic LVOT obstruction in HCM. Sudden ventricular arrhythmia from the hypertrophied, fibrotic septum can be lethal.

EXERCISE & AGING CVS

4 CARDS

55 Net change in SVR, SBP, DBP, and PP during dynamic aerobic exercise?

SVR **falls** (muscle vasodilation outweighs splanchnic constriction). SBP rises, DBP unchanged or slightly falls, PP widens.

56 Aerobic training vs isometric training — which produces concentric vs eccentric LVH?

Aerobic / volume (running, swimming) → **eccentric** LVH (sarcomeres in series). **Isometric / pressure** (weightlifting) → **concentric** LVH.

57 Mechanism of isolated systolic hypertension in elderly patients?

Progressive loss of aortic elastin → ↓ aortic compliance → the same stroke volume creates a larger pressure swing → high SBP, normal DBP, wide PP.

58 Identity and origin of lipofuscin?

Yellow-brown intracytoplasmic "wear-and-tear" pigment from lipid peroxidation of organelle membranes. Accumulates with age in non-dividing cells (heart, liver, brain).